

RCS-VRF PROJECT VERIFICATION & CERTIFICATION DESIGN

(v0.4.1 — math layer closed)

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Problem Statement

LEG 5 IN THE PIPELINE

RCS-VRF → Quantum Proof of Useful Work subnet that emits verifiable quantum randomness.

- ▶ LEG 1: Initial seed from public, bias-resistant sources. (*Mehdi's deck*)
- ▶ LEG 2: Circuit-generation parameters precommitted before seed is used.
- ▶ LEG 3: Seed deterministically generates RCS circuit.
- ▶ LEG 4: Quantum nodes produce samples + XEB evidence.
- ▶ **LEG 5: Classical validators check timing, XEB threshold, and certify entropy. (*this deck*)**
- ▶ LEG 6: Toeplitz/Trevisan extractor turns accepted samples into randomness output.
- ▶ LEG 7: Randomness anchored into a QPoUW block and served to consumers.

Deck boundary. This deck focuses on how accepted samples become certified randomness (LEG 5).

WHAT LEG 5 MUST DO

The verifier sits between the (untrusted) quantum server and the customer.

Given a batch of M samples from M random circuits, the verifier must:

- ▶ Check that samples plausibly came from genuine quantum sampling.
- ▶ Quantify how much certified randomness is in the batch.
- ▶ Decide accept-or-abort. If accept, pass the entropy budget downstream.

Without Leg 5, the protocol has no certification. It is the step that turns “the server sent us some bits” into “these bits contain ℓ bits of provably-quantum randomness.”

LEG 5 REQUIREMENTS

- ▶ **Adversary-aware**: secure even when the quantum server is dishonest.
- ▶ **Quantifiable**: produces a numerical bound on certified entropy, not just a pass/fail flag.
- ▶ **Classically verifiable**: the verifier uses only a classical computer.
- ▶ **Auditable**: any third party can re-derive the bound from public inputs in the audit log.
- ▶ **Timing-aware**: the response deadline is part of the security argument, not just a fairness rule.

Approaches Considered

OPTION A — AARONSON–HUNG (2023) STRUCTURE

Setup: ONE random circuit per round, MANY shots on that single circuit.

Verification: F_{XEB} averaged over the shots.

The structural challenge: the many shots within a round are *correlated* through the shared circuit — not independent samples. A cheater's amortized strategy (simulate the circuit's output distribution once, then sample many times from that distribution) reduces per-shot cost relative to per-circuit cost.

The AH bound *does* handle this correctly — it uses the **Entropy Accumulation Theorem (EAT)** to account for the correlation across shots. The proof works. But the EAT machinery yields a *looser* certified-entropy bound for the same parameters than i.i.d. accumulation would.

Our v0 through v0.3a implemented this structure.

OPTION B — LIU ET AL. (2025) STRUCTURE

Setup: MANY *different* circuits ($M = 30,010$ in Liu's paper), exactly ONE shot per circuit.

Verification: F_{XEB} averaged over an audit subset (m of M).

Key insight: one shot per distinct circuit makes the rounds *independent* — there is no within-round correlation to handle. Each round is a fresh, unrelated experiment.

This enables a cleaner security argument via standard i.i.d. concentration bounds (no EAT machinery needed) and yields a *tighter* certified-entropy bound for the same parameters.

We switched to this structure in v0.4.

WHY WE SWITCHED

	AH structure (v0–v0.3a)	Liu structure (v0.4 onward)
Circuits per round	1	M distinct
Shots per circuit	many	exactly 1
Within-round structure	correlated shots	independent rounds
Security argument	EAT (handles correlation)	i.i.d. concentration
Bound tightness	looser	tighter

Important precision: the AH bound is *correct*, not broken. The switch isn't a bug fix — it's a structural choice that yields a tighter bound and a simpler proof. “Rounds” under Liu's structure also become a meaningful unit: $Q_{\min}$ is a number of distinct quantum circuits that had to be honest, not a number of shots.

Verification Chain

HAAR-RANDOM CIRCUITS — WHERE THE RANDOMNESS LIVES

Haar measure = the unique uniform distribution over unitary matrices.

- ▶ A *Haar-random* unitary U on n qubits is sampled uniformly from $U(2^n)$.
- ▶ Output state on $|0\rangle^n$: $|\psi\rangle = U|0\rangle^n$, a uniformly random pure state.

Why this matters for certification: a Haar-random unitary destroys all structure.

- ▶ No subset of bitstrings is preferentially favored *ex ante*.
- ▶ No classical shortcut can exploit special symmetries in the circuit.
- ▶ Output statistics are universal and analytically tractable.

Intuition: the “perfect shuffle” of quantum operations — nothing left of the initial state’s structure.

The catch: a truly Haar-random unitary requires exponentially many gates to implement. Building one exactly is infeasible. We need an approximation.

t -DESIGNS AND RCS — THE ENGINEERING TRICK

A **unitary t -design** is a finite ensemble of unitaries that *reproduces the first t moments of the Haar distribution*.

- ▶ Smaller, finite, and efficiently sampleable.
- ▶ For our purposes (F_{XEB} , collision entropy) we only need the **second moment**.
- ▶ So a 2-design is enough — exact Haar randomness is overkill.

Random Circuit Sampling (RCS): a recipe for approximate t -designs using shallow random circuits.

- ▶ Alternating layers of Haar-random $\text{SU}(2)$ single-qubit gates and entangling two-qubit gates.
- ▶ After $O(n)$ layers, the resulting unitary is statistically indistinguishable from Haar to second-moment tests.

This is the engineering bridge: “too hard to build Haar” $\rightarrow$ “a deep enough RCS circuit is Haar-indistinguishable for our purposes.”

HOW CIRCUITS ARE ACTUALLY CONSTRUCTED

Deterministic from the seed via Leg 2 + Leg 3:

$$\text{seed} \xrightarrow{\text{PRF}} \text{byte stream} \xrightarrow{\text{construction}} \{C_1, C_2, \dots, C_M\}$$

Per-circuit recipe (Liu's setup, repeated ~ 24 layers):

- ▶ **Single-qubit layer:** consume bytes to pick a Clifford rotation for each of the n qubits (from a fixed set of ~ 24 rotations).
- ▶ **Two-qubit layer:** consume bytes to pick a random perfect matching of the n qubits into $n/2$ pairs; apply the native two-qubit gate to each pair.

Byte budget: ~ 1 KB per circuit at $n = 56$, depth 24. With $M = 30,010$ circuits per Quip-round, total ~ 30 MB — which is exactly why Leg 2 needs cryptographic expansion of the small Leg 1 seed.

Reproducibility: same seed $\rightarrow$ same byte stream $\rightarrow$ same circuits. Any verifier can re-derive the circuits from the published seed and check independently.

The seed selects circuits; the construction recipe ensures the family is an approximate t -design. Both pieces are needed for Haar-like behavior.

THE PORTER–THOMAS DISTRIBUTION — THE PHYSICS PAYOFF

Claim: for a Haar-random (or sufficiently deep RCS) n -qubit circuit, the output probabilities $p(x) = |\langle x|U|0\rangle|^2$ follow:

$$\Pr[N \cdot p(x) \in [s, s + ds]] = e^{-s} ds, \quad N = 2^n.$$

The rescaled probabilities are *exponentially distributed*. This is the **Porter-Thomas distribution** (Porter & Thomas, 1956; originally nuclear physics).

The magic moment: the second moment of p under Porter-Thomas is:

$$\sum_x \mathbb{E}[p(x)^2] = \frac{2}{N}.$$

Why this is magic:

- ▶ It gives an *exact, universal* formula for the protocol's per-round entropy.
- ▶ It is *the same for every circuit family* that's a 2-design.
- ▶ It is *checkable from samples* — the F_{XEB} score statistically tests this property.

This single number, $2/N$, is what makes the whole certification chain possible.

WHAT IS “CERTIFIED MIN-ENTROPY”?

Min-entropy measures *guaranteed unpredictability* (in bits).

$$H_{\min}(X) = -\log_2\left(\max_x \Pr[X = x]\right)$$

If the best guess at X succeeds with probability p , then X has $-\log_2 p$ bits of min-entropy.
Tighter and more conservative than Shannon entropy.

“Certified” adds two requirements:

- ▶ The bound holds *against a worst-case adversary* in our model — not just on average over random inputs.
- ▶ The bound is *provable from public evidence* (the audit log), not merely asserted by the server.

Concretely: certified min-entropy $H_{\min}^{\text{cert}} = X$ bits means

“even the most capable adversary in our threat model, looking at all public protocol data, cannot guess the raw output with probability greater than 2^{-X} .”

This number is the input budget passed to Leg 6, which then extracts $\ell \leq X$ near-uniform bits from the raw samples via Toeplitz/LHL.

THE CHAIN AT A GLANCE

raw quantum samples

↓ F_{XEB} on audit subset

score (and timing check)

↓ Liu adversary model (Lemma 2)

$Q_{\min}$ (minimum forced-quantum rounds)

↓ multiply by $n - 1$ (collision entropy)

↓ subtract smoothing margin $\log_2(1/\varepsilon_s)$

$H_{\min}^{\text{cert}}$ (certified bits to Leg 6)

Each arrow is one specific formula. Next slides walk each link.

STEP 1 — F_{XEB} COMPUTATION

For an audit subset of m circuits, the verifier classically simulates each and computes:

$$F_{\text{XEB}} = \frac{N}{m} \sum_{i=1}^m p_{C_i}(x_i) - 1$$

Four expected cases:

- ▶ **Uniform noise / blind spoof:** $F_{\text{XEB}} \approx 0$
- ▶ **Real noisy quantum machine:** $F_{\text{XEB}} \approx \phi$ (fidelity, in $(0, 1)$)
- ▶ **Perfect quantum:** $F_{\text{XEB}} = 1$
- ▶ **Cherry-picking spoof:** $F_{\text{XEB}} \gg 1$

Liu et al.'s acceptance threshold: $\chi = 0.3$.

STEP 2 — ACCEPTANCE CHECK (A DELIBERATE DEVIATION FROM LIU)

Liu uses a one-sided check: accept iff $F_{\text{XEB}} \geq \chi$.

Their argument: a compute-rational adversary will not score $\gg 1$ (that would require *more* honest quantum work, not less). So no upper bound is needed.

Our v0.1+ uses a two-sided check:

$$\chi_{\text{low}} \leq F_{\text{XEB}} \leq \chi_{\text{high}}$$

Strictly more conservative than Liu requires. Catches a different threat class:

- ▶ Wasteful or non-rational adversaries.
- ▶ Pure heavy-output generation attacks.
- ▶ Adversaries who circumvent the compute-budget assumption.

Defence-in-depth choice. The lower bound is load-bearing; the upper bound is our addition.

STEP 3 — TIMING (THE SECOND HALF OF THE PROOF)

F_{XEB} alone proves only *correctness*, not *quantum origin*. A patient classical attacker with unbounded time could simulate every circuit, score perfectly, and pass.

The response deadline closes the classical-faking route:

- ▶ Genuine quantum machine clears the deadline naturally (seconds).
- ▶ Classical simulator of $n = 56$ circuits takes hours per circuit.

$F_{\text{XEB}} \checkmark + \text{timing} \checkmark = \text{proof of quantumness.}$
Neither alone is sufficient.

Status in our implementation: timing currently stubbed. Recorded but not enforced. The math chain assumes enforcement; engineering is owed.

STEP 4 — THE ADVERSARY MODEL

Liu's “restricted realistic near-term adversary” is given:

- ▶ Perfect-fidelity quantum hardware (best-case quantum capability).
- ▶ 1.1×10^{18} FLOPS classical compute (current top supercomputers).
- ▶ Best-known classical attacks on F_{XEB} .
- ▶ Same per-circuit deadline as the honest server.

Strategy: minimize honest quantum work while still passing F_{XEB} .

Within the deadline, the adversary's classical faking budget is at most:

$$\Phi_A \approx \frac{A \cdot t_{\text{threshold}}}{B}$$

where B is the cost of faking one circuit's F_{XEB} contribution.

Goal: bound how few quantum rounds the adversary could have gotten away with.

STEP 5 — $Q_{\min}$ DERIVATION

$$Q_{\min} = \min \{ Q : \varepsilon_{\text{adv}}(Q, \chi) \geq \varepsilon_{\text{sou}} \}$$

In words: the smallest number of honest quantum rounds at which the adversary's probability of passing $F_{\text{XEB}} \geq \chi$ still exceeds the soundness threshold ε_{sou} .

- ▶ Computed via hypergeometric-tail bound with closed-form Chernoff δ .
- ▶ Bridges F_{XEB} (a statistical observation) and certified min-entropy (a cryptographic guarantee).
- ▶ Same number as the adversary's threshold and our certification's floor.

For Liu's paper parameters: $Q_{\min} = 1297$ (out of $M = 30,010$).

Reproduced exactly by our calibrated implementation (`leg5_qmin_liu.py`).

STEP 6 — PER-ROUND ENTROPY: $n - 1$

The collision (2-Rényi) entropy of the Porter-Thomas distribution:

$$H_2 = -\log_2\left(\sum_x p(x)^2\right) = -\log_2\left(\frac{2}{N}\right) = \log_2 N - 1 = n - 1$$

For Liu's $n = 56$ qubits: **55 bits** of certified entropy per honest quantum round.

- ▶ Derived from first principles via the Porter-Thomas second moment.
- ▶ No convention freedom — forced by the distribution.
- ▶ Same across Liu, Aaronson-Hung, and any RCS-based certified-randomness protocol.

This is the most cleanly nailed-down piece of the chain.

STEP 7 — CERTIFIED MIN-ENTROPY (LIU COROLLARY 7)

$$H_{\min}^{\text{cert}} = Q_{\min} \cdot (n - 1) - \log_2\left(\frac{1}{\varepsilon_{\text{sou}}}\right) - 2$$

For Liu's parameters ($Q_{\min} = 1297$, $n = 56$, $\varepsilon_{\text{sou}} = 10^{-6}$):

$$H_{\min}^{\text{cert}} = 1297 \cdot 55 - \log_2(10^6) - 2 = 71,335 - 19.93 - 2 = \mathbf{71,313.07 \text{ bits}}$$

This is the entropy budget passed to Leg 6 (Toeplitz extraction).

Reproduced exactly by our verification harness against Liu's experimental data.

STEP 8 — TOEPLITZ / LHL EXTRACTION (LIU COROLLARY 7)

$$\ell \leq Q_{\min} \cdot (n - 1) - 3 \log_2 \left(\frac{1}{\varepsilon_{\text{sou}}} \right) - 2$$

Why a separate step: certified min-entropy lives inside the raw quantum samples, but the samples themselves are not uniformly distributed — they carry structure from the circuit. The **Leftover Hash Lemma** guarantees that hashing a high-min-entropy source through a 2-universal hash (e.g., a **Toeplitz matrix**) yields output statistically close to uniform.

For Liu's parameters:

$$\ell = 1297 \cdot 55 - 3 \log_2(10^6) - 2 = 71,335 - 59.79 - 2 = \mathbf{71,273.21} \text{ bits}$$

Extraction overhead: 40 bits lost relative to certified min-entropy — the cost of going from “entropy exists somewhere” to “uniformly random bits.”

Final deliverable: uniformly distributed bits, ε_{sou} -close to uniform, conditionally independent of adversary side-information.

Open Items

CLOSED ITEM 1 — $Q_{\min}$ CALIBRATION (v0.4.1)

Reconciled against Liu's Zenodo reference notebook:

- ▶ Adversary parameters identified from
Table2-bounds-on-extractable-entropy.ipynb: $\text{eff} = 0.5$,
 $A = 4 \times \text{FRONTIER}_{\text{theoretical}} = 8 \times 10^{18}$ FLOPS.
- ▶ Closed-form Chernoff δ replaces numerical grid.
- ▶ Half-budget threshold split: $\varepsilon_{\text{sou}}/2$ on each tail (instead of $4\varepsilon_s$ combined).

Calibrated module: `leg5_qmin_liu.py`.

Reproduces Liu exactly:

- ▶ $Q_{\min} = 1297$ (paper: 1297) ✓
- ▶ $H_{\min}^{\text{cert}} = 71,313.07$ bits (paper: 71,313) ✓
- ▶ Extractable $\ell = 71,273.21$ bits (paper: 71,273) ✓

Previous `xfail` retired. Regression test `test_qmin_calibrated.py`: 5 passing tests.

TRIPLE-VERIFIED — THREE INDEPENDENT TEST PATHS

1. Algorithm correctness (`test_qmin_calibrated.py`)

- ▶ 5 regression tests: exact paper reproduction, protocol breakdown at strong adversary, monotonicity in soundness and adversary compute.

2. Real-data correctness (`verify_against_liu_data.py`)

- ▶ Full chain on Liu's published `aggregated_probs.npy` (1,522 precomputed probabilities from Quantinuum H2-1).
- ▶ Reproduces $F_{\text{XEB}} = 0.32$, $Q_{\text{min}} = 1297$, $H = 71,313.07$, $\ell = 71,273.21$. All match to the digit.

3. Formula correctness (`test_fxeb_limiting_cases.py`)

- ▶ 7 limiting-case tests: perfect quantum ($\phi = 1$), pure noise ($\phi = 0$), mixed fidelity, size-biasing isolation, worst-case argmax adversary, single-shot structure, LLN concentration scaling.

Three independent angles: most failure modes are caught by at least two paths.

OPEN ITEM 2 — TIMING ENFORCEMENT

Today:

- ▶ Timestamps are recorded in the audit log.
- ▶ No deadline is enforced in code.

Why this matters: the entire proof of quantumness rests on the response deadline. Without enforcement, a patient classical attacker could still pass.

Possible directions:

- ▶ Hash-graph propagation as a timing oracle (*Mehdi's Leg 1 deck flagged this*).
- ▶ External time anchor via drand pulse ordering.
- ▶ Cryptographic timestamping with the audit-log hash chain.

Status: known engineering stub; design TBD with Mehdi.

TAKEAWAY

Math layer closed (v0.4.1).

- ▶ Leg 5 implements Liu et al.'s certification chain faithfully (Corollary 7).
- ▶ Per-round collision entropy $n - 1$ is exact, derived from Porter-Thomas.
- ▶ Protocol structure switched to Liu (many circuits, one shot each) in v0.4.
- ▶ $Q_{\min}$ calibration completed; reproduces Liu's $Q_{\min} = 1297$ and downstream numbers exactly.
- ▶ Triple-verified via three independent test paths (algorithm, real-data, formula).

One deliberate deviation from Liu, retained:

- ▶ Two-sided F_{XEB} check (defence-in-depth; strictly more conservative).

The math chain is closed end-to-end. Reproduces every published number Liu reports.

Questions